

Enhancing Autonomous Driving Systems: Integrated Lane and Object Detection Using YOLOv3 and OpenCV

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Abstract—Autonomous driving systems require reliable perception capabilities for safe and efficient navigation, particularly in lane and object detection. This study presents a real-time detection framework that integrates deep learning with traditional computer vision techniques. Lane detection is achieved using the Canny edge detection algorithm, Hough transform for line extraction, and color thresholding in the HSV color space. For object detection, the YOLOv3 model, pre-trained on the COCO dataset, is used to recognize vehicles, pedestrians, and so on, road obstacles. The application of these practices together offers lane- while ensuring accuracy and traffic awareness, producing real-time warnings like "Stay On The Lane" or "Go Slow, Traffic." The proposed technique provides high accuracy of detection, computational efficiency, and stability under various environmental conditions. Its use also includes advanced driver assistance systems (ADAS), autonomous driving navigation, and smart traffic management, leading to better road traffic safety and regulation.

Index Terms—YOLOv3, Hough Transform, Canny Edge Detection, Lane Detection, Computer Vision, Object Detection

I. INTRODUCTION

Autonomous driving is revolutionizing transportation by advancing technologies such as advanced driver assistance systems (ADAS) [1], smart traffic networks, and autonomous vehicles. Imagine driving through a busy city center during rush hour: as you focus on your destination, your vehicle uses its systems to identify traffic lights, pedestrians, and lane markings in real time, creating a safe and smooth ride. These systems aim to reduce accidents, optimize traffic flow, and enable easier and safer mobility for everyone.

However, the subtleties of real-world driving, such as varying illumination, unpredictable weather, and inconsistent behavior by other vehicles, pose significant challenges. For example, a sunset road can confuse systems relying on color-based approaches, while fog may obscure lane markings, leading to detection failure. Solutions capable of adapting to diverse and dynamic environments are essential to overcome these issues.

Classic lane and object detection methods generally perform poorly under such conditions. For example, basic color thresholding may not differentiate lane lines in shadow, and edge detection routines may misinterpret cracks on the road as

lane edges. Similarly, early object detection systems lacked the accuracy and efficiency necessary for real-time autonomous vehicle requirements. These limitations highlight the need for end-to-end solutions that integrate traditional computer vision techniques with contemporary deep learning methods.

To meet these challenges, this paper proposes a real-time lane and object detection system that combines conventional computer vision techniques with the YOLOv3 deep learning architecture. Consider the scenario of driving on a mountain road: the system converts RGB camera images to the HSV color space, allowing precise lane marking segmentation under quickly changing sunlight and shade. Canny edge detection is applied to detect significant edges, followed by the Hough Transform to extract lane boundaries optimized for performance, ensuring robust operation under varying illumination conditions.

For object detection, consider a busy city street where pedestrians may step onto the road unexpectedly or a car may be parked across a lane. Using the YOLOv3 model trained on the COCO dataset [2], the system detects objects such as cars, pedestrians, and traffic signs [3] with high speed and accuracy. OpenCV [4] is used to seamlessly combine these detection modules. The system overlays lane detection information onto object detection outputs to generate actionable commands such as "Stay on the Lane" or "Traffic Ahead: Slow Down." These features improve situational awareness, enabling the system to function robustly in real-world autonomous driving scenarios.

Through the integration of deep learning and traditional computer vision, this solution provides a scalable, real-time approach to autonomous navigation, addressing key challenges in lane and object detection in real-world environments.

II. LITERATURE REVIEW

Autonomous driving technology has advanced rapidly over the last few years with a considerable focus on the core perception tasks of object detection and lane detection, both foundational for safe driving. Both tasks facilitate the accurate detection of road edges, obstacles, and other traffic participants. This chapter assesses the state of the art in object detection and lane detection and situates these approaches in the context of our proposed approach.

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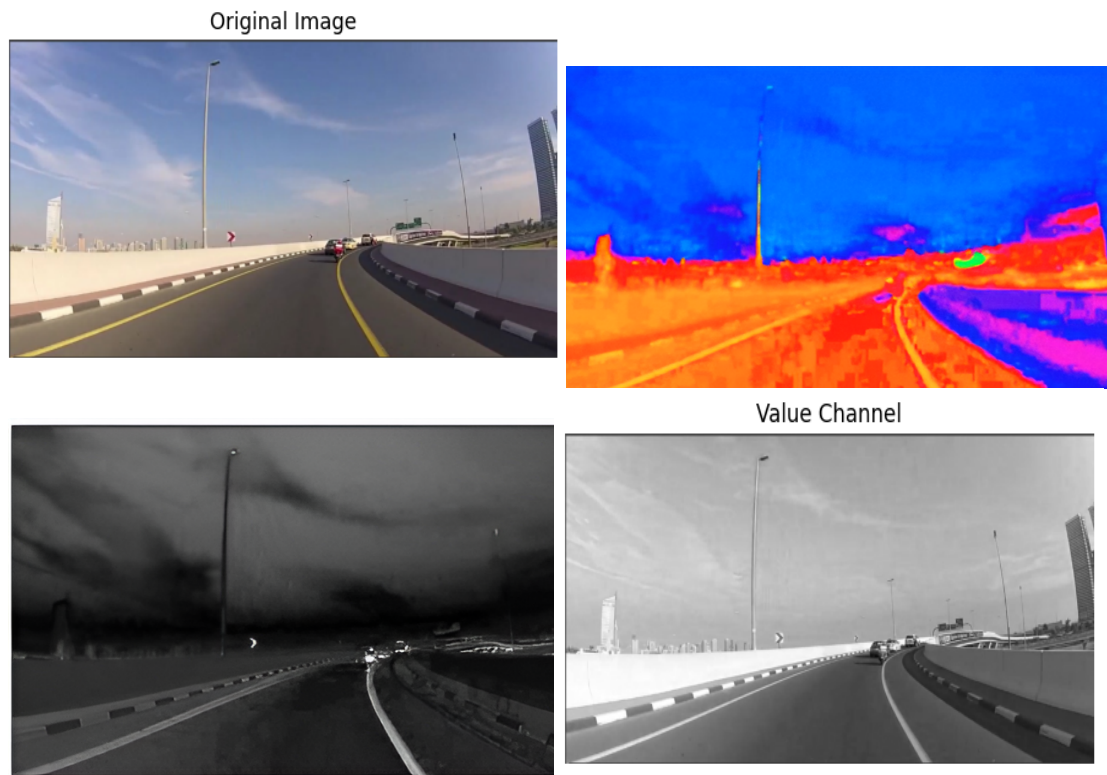


Fig. 1. Visual representation of the original image and its HSV channels. (Upper left: Original image; Upper right: Hue channel; Lower left: Saturation channel; Lower right: Value channel.)

A. Lane Detection

Lane detection is the foundation of autonomous driving, as it enables vehicles to recognize road edges and remain in the right lane. Early methods were primarily built upon conventional computer vision algorithms. For instance, the Canny edge detector [5] was commonly used to enhance lane markings, and the Hough Transform [6] was utilized to identify linear lane shapes. These algorithms were computationally efficient and performed well under ideal circumstances. [7] However, they could not cope with real-world conditions, such as curved roads, occlusions, or low lighting, and were thus limited in their application.

To counter these challenges, researchers proposed color-based segmentation methods in the HSV color space to enhance lane detection under different lighting conditions. Although invariant to illumination changes, these methods were less robust in cases of worn-off lane markings, heavy traffic, or occluded road lines.

The advent of deep learning has transformed lane detection by allowing models to learn directly from images. Convolutional Neural Networks (CNNs) have demonstrated great performance in detecting complex lane patterns, such as dashed or curved lanes. For instance, LaneNet [8] and Spatial CNN (SCNN) [9] have significantly improved lane detection performance in different scenarios. However, these methods often require significant computational resources, making real-time execution on embedded systems challenging.

Here, we address these limitations by integrating conventional computer vision methods (HSV color thresholding, Canny edge detection, and Hough Transform) with object detection based on deep learning. The proposed hybrid method exploits the advantages of both approaches to provide a scalable, real-time solution tuned for embedded devices.

B. Object Detection

Object detection plays a vital role in autonomous driving by identifying vehicles, pedestrians, and road signs. Early object detection methods relied on hand-designed features such as Haar cascades and Histogram of Oriented Gradients (HOG). Though such approaches worked adequately in well-managed environments, they were not adaptable to different road conditions and performed poorly in detecting objects with high precision.

The advent of deep learning has greatly improved object detection accuracy and efficiency. Models like Faster R-CNN [10], Single Shot Detector (SSD) [11], and You Only Look Once (YOLO) [12] have demonstrated significant advancements in real-time detection. Among them, YOLOv3 [13] is remarkable for achieving an optimal balance between speed and accuracy. By partitioning an image into grid blocks and predicting object bounding boxes and categories, YOLOv3 can perform rapid and precise object identification, making it particularly suitable for autonomous vehicle applications.

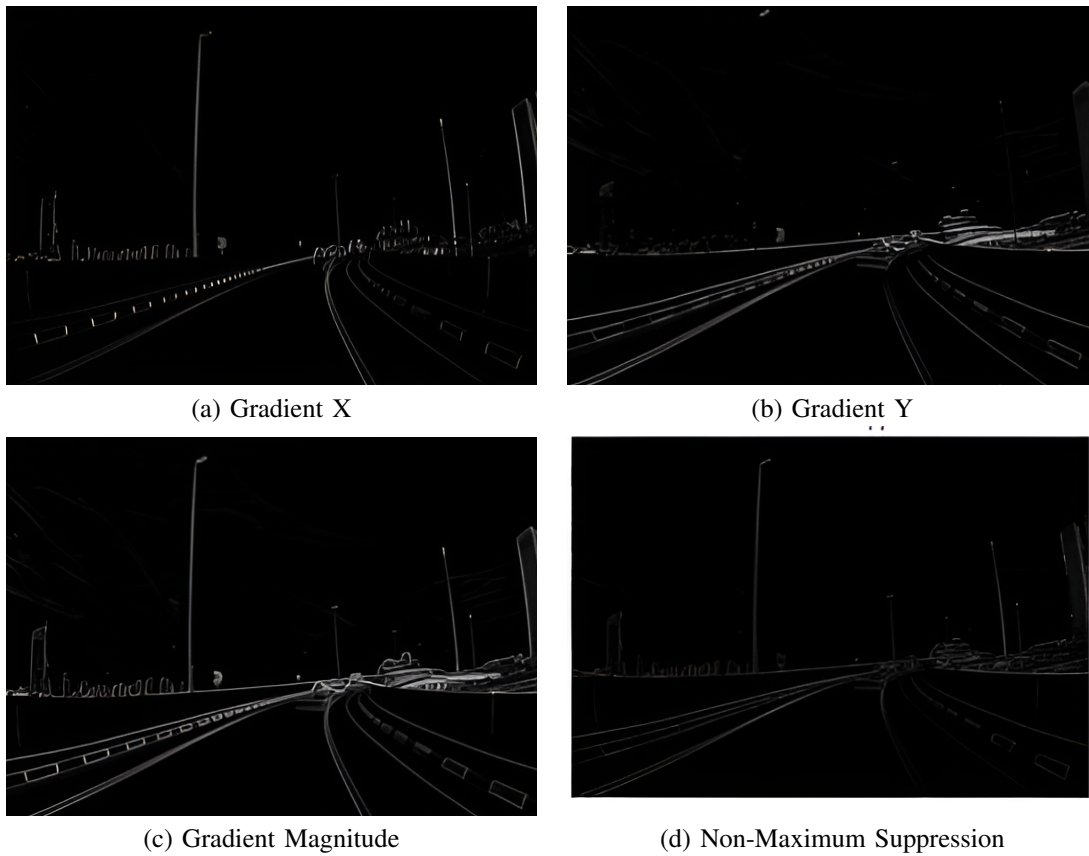


Fig. 2. Visualization of gradient computation and edge refinement steps. (a) Horizontal gradient (G_x); (b) Vertical gradient (G_y); (c) Gradient magnitude (G); (d) Non-Maximum Suppression applied to retain the strongest edges.

Despite these advancements, deep learning-based object detection remains computationally expensive and necessitates large annotated datasets for effective training. To address these challenges, researchers have explored optimization methods like transfer learning and model quantization, which enable efficient deployment on resource-constrained hardware.

C. Integration of Lane and Object Detection

Lane and object detection are essential for autonomous vehicles, forming an integrated perception system that improves environmental understanding and decision-making. Lane detection aids path planning, while object detection ensures hazard awareness for safer maneuvering.

Recent research explores multi-task learning frameworks using a shared CNN backbone or hierarchical methods where lane detection refines object detection for better positioning.

A major challenge is achieving real-time performance without losing accuracy. Hybrid approaches combining traditional computer vision for lane detection with deep learning for object detection provide a cost-effective, accurate solution suitable for embedded autonomous systems.

III. PROPOSED METHODOLOGY

The proposed system combines classical computer vision techniques with deep learning for real-time lane and object detection in autonomous vehicles. Each video frame undergoes

preprocessing to enhance image quality and reduce noise using Gaussian blur before conversion to the HSV color space for robust segmentation of lane markings. Edge detection is performed using Sobel filters, followed by non-maximum suppression [14] and double thresholding to generate an edge map. The Hough Line Transform [6] is then applied to identify lane boundaries.

Simultaneously, the frame is resized, normalized, and converted into a blob format for processing by the YOLOv3 [13] deep learning model. This model generates bounding boxes, confidence scores, and class probabilities at multiple scales, enabling detection of objects such as vehicles and pedestrians.

The outputs from both pipelines are combined and overlaid onto the original frame, providing real-time visual feedback and navigational cues for safe operation. The complete detection pipeline consists of four main stages: (1) Preprocessing, (2) Lane Detection, (3) Line Detection, and (4) Object Detection, which are discussed in detail in the following subsections

A. Preprocessing

Preprocessing is crucial for enhancing the quality of input video frames and removing noise before further analysis. In our framework, each frame is first refined using a Gaussian blur to remove high-frequency noise and smooth the image. Gaussian blur is applied by convolving the image with a 5×5

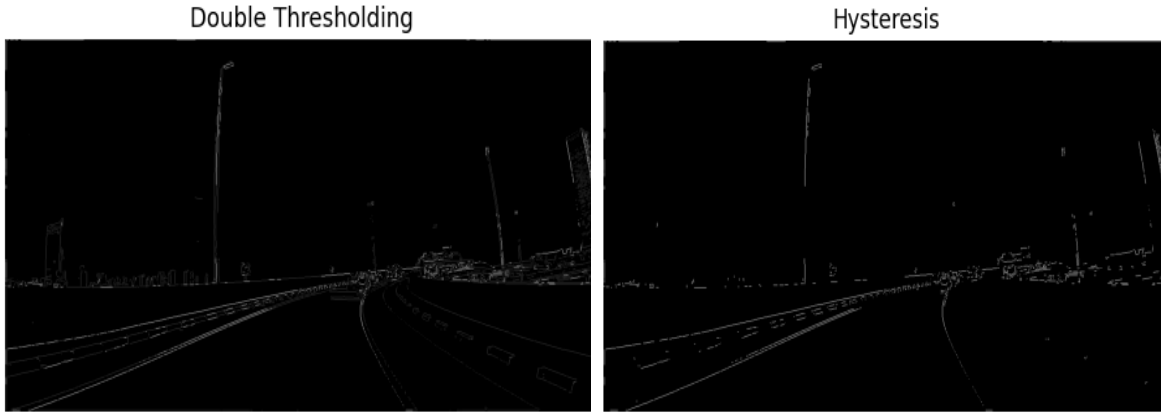


Fig. 3. Edge refinement in lane detection. (Left: Double Thresholding classifies edges as strong or weak; Right: Hysteresis ensures continuity by preserving significant edges and discarding isolated weak edges.)

Gaussian kernel with a standard deviation of 0. The Gaussian kernel is defined as:

$$G(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right) \quad (1)$$

Here, $G(x, y)$ represents the kernel value at a pixel located at (x, y) , σ is the standard deviation controlling the spread of the Gaussian distribution, and the constant $\frac{1}{2\pi\sigma^2}$ normalizes the kernel.

B. Lane Detection

In our system, each input frame undergoes preprocessing with a Gaussian blur to reduce noise, followed by conversion from the BGR to HSV color space to isolate lane markings based on color. Sobel filters compute the gradient magnitude and direction, enhancing edge detection. Non-Maximum Suppression (NMS) [14] refines these edges by retaining only local maxima, while double thresholding and hysteresis further classify and preserve edges most likely to represent true lane boundaries. [5]

- 1) **HSV Color Space Conversion:** The input frame is converted to HSV color space to enhance lane marking segmentation under varying lighting conditions. A specific hue range is used to detect yellow lane markings:

$$\text{Lower Yellow} = [18, 94, 140],$$

$$\text{Upper Yellow} = [48, 255, 255]$$

This step produces a binary mask highlighting areas within the defined yellow range.

- 2) **Gradient Calculation:** To detect lane boundaries, Sobel filters compute intensity gradients in both horizontal (G_x) and vertical (G_y) directions. The gradient magnitude and direction are given by:

$$G = \sqrt{G_x^2 + G_y^2}, \quad \theta = \tan^{-1}\left(\frac{G_y}{G_x}\right) \quad (2)$$

where G_x and G_y denote intensity changes in respective directions, with G representing edge strength and θ indicating orientation. The computed gradient components are illustrated in Figure 2 (a-c).

- 3) **Non-Maximum Suppression (NMS):** To refine detected edges, NMS is applied, retaining only the strongest edge responses by suppressing non-maximum pixels along the gradient direction:

$$G(x, y) = \begin{cases} G(x, y), & \text{if } G(x, y) > \max(G_{nb1}, G_{nb2}) \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

As shown in Figure 2 (d), NMS ensures that only the most prominent lane edges are retained.

- 4) **Double Thresholding and Edge Tracking:** To further refine lane edges, double thresholding is applied with a high threshold ($T_H = 150$) and a low threshold ($T_L = 74$), classifying edges as strong or weak. Weak edges connected to strong edges are retained through hysteresis, enhancing lane continuity (Figure 3).

The results in Figure 2 indicate that gradient computation successfully highlights lane edges, while Non-Maximum Suppression further filters out irrelevant edges. Figure 3 shows that applying thresholding and hysteresis suppresses weak edges while maintaining lane continuity, resulting in a robust lane detection system.

C. Line Detection

The refined edges from the lane detection stage are processed using the Hough Line Transform [6], a voting-based technique that identifies straight lines by transforming edge pixels into a parameter space. Instead of using the standard Cartesian equation $y = mx + c$, the Hough Transform represents lines in polar form as:

$$r = x \cos \theta + y \sin \theta \quad (4)$$

where r is the perpendicular distance from the origin to the line, θ is the angle between the x-axis and the line's normal,

and (x, y) are the coordinates of any point on the line. This parameterization enables efficient detection of lines, regardless of their orientation.

The Hough Transform maps each edge pixel (x, y) to multiple (r, θ) pairs, accumulating votes in a discretized parameter space known as the Hough accumulator. Peaks in this accumulator indicate the presence of prominent lines.

This method efficiently identifies lane markings by detecting strong, continuous lines from edge maps. The detected lines are superimposed on the original frame to visualize lane boundaries. By adaptively adjusting both the voting threshold and maximum line gap parameters, the algorithm achieves an optimal balance between detection sensitivity and system robustness, ensuring reliable lane identification under varying road conditions.

D. Object Detection

Object detection is implemented using the YOLOv3 deep learning model, which excels in real-time applications. The process consists of several key steps:

Preprocessing

Frames undergo preprocessing before being fed to the YOLOv3 model to ensure compatibility and enhance detection accuracy:

- **Resizing:** Input frames are resized to 416×416 pixels to match YOLOv3's input dimensions
- **Normalization:** Pixel values are normalized to the range $[0, 1]$ to facilitate network convergence
- **Blob Conversion:** Processed frames are converted into blobs using `cv2.dnn.blobFromImage` with: `leftmargin=*`
 - Scale factor of $\frac{1}{255}$
 - Aspect ratio preservation
 - Red and Blue channel swapping (BGR to RGB)

1) *YOLOv3 Architecture:* YOLOv3 [13] (You Only Look Once, Version 3) represents a state-of-the-art object detection framework that significantly outperforms its predecessors YOLOv1 and YOLOv2 [15]. Designed specifically for real-time applications, YOLOv3 employs a single-pass detection approach that simultaneously predicts both class probabilities and bounding box coordinates. This architecture achieves an optimal balance between speed and accuracy compared to conventional two-stage detectors like Faster R-CNN.

The network employs Darknet-53 as its backbone, a deep 53-layer convolutional neural network with residual connections for enhanced feature extraction. YOLOv3 uses multi-scale detection at three different resolutions— 13×13 , 26×26 , and 52×52 —allowing it to detect objects of various sizes effectively. Predefined anchor boxes, derived via K-means clustering, provide priors for bounding box predictions, refined during training. The detection head outputs bounding boxes, confidence scores, and class probabilities. The architecture is summarized in Table I.

E. Performance Comparison

To evaluate the effectiveness of our proposed lane and object detection system, we compare its performance with conventional methods based on accuracy and processing speed (FPS). Our model is pretrained on the COCO dataset [2], which contains a wide range of annotated images, including urban and highway lane scenarios. The dataset aids in learning robust lane features under varying lighting and weather conditions. The pretrained weights are fine-tuned on a custom lane detection dataset consisting of 10,000 annotated lane images, improving domain-specific accuracy.

F. Evaluation Metrics

The assessment procedure of the proposed system is designed to verify its performance and accuracy under real-time autonomous vehicle conditions. The following critical measurements are utilized:

- **Accuracy (%):** The percentage of correctly identified lane pixels or objects with respect to ground truth annotations. This metric provides a quantitative measure of how well the system approximates the labeled dataset.
 - **Validation Process:** Validation was conducted on a labeled dataset with manually annotated ground truth for lanes and objects. Accuracy is calculated by evaluating the detected lanes or bounding boxes against the manually annotated ground truth. The Intersection over Union (IoU) metric is used for object detection, while pixel-wise matching is employed for lane detection.
- **Frames Per Second (FPS):** FPS measures the real-time processing rate of the system, representing the number of video frames processed per second.
 - **Validation Process:** FPS was quantified by running the system on a standard hardware configuration, specifically an Intel i7 processor and an NVIDIA GTX 1080 GPU. The average time taken to process a batch of video frames was recorded to calculate the FPS.

G. Quantitative Results

The YOLOv3 model was initialized using pretrained weights from the COCO dataset and then fine-tuned on a specific dataset for detecting objects and lanes. This fine-tuning process enables the model to generalize to the unique challenges of autonomous driving environments, such as varying lighting conditions and diverse road textures.

IV. RESULTS AND ANALYSIS

The performance of the proposed system was evaluated in real-world driving scenarios to analyze its effectiveness in lane detection and object detection. The results demonstrate that the system accurately identifies lane boundaries and road objects, providing timely warnings for autonomous navigation.

TABLE I
YOLOV3 ARCHITECTURE SUMMARY

Component	Functionality
Input Image	Resized, normalized
Backbone (Darknet-53)	Feature extraction via a 53-layer CNN with residual blocks
Multi-Scale Detection	Detection at 13×13, 26×26, and 52×52 grids
Anchor Boxes	Predefined priors for refining bounding box predictions
Detection Head	Outputs bounding boxes, confidence scores, and class probabilities

TABLE II
IMPLEMENTATION PARAMETERS FOR HOUGH LINE TRANSFORM

Parameter	Value
Distance resolution	1 pixel
Angular resolution	1° ($\pi/180$ radians)
Vote threshold	50
Maximum line gap	50 pixels

A. Lane Detection Performance

The lane detection module exhibited high accuracy in detecting lane markings, particularly under standard road conditions with clear and visible lanes. The system successfully identified lane boundaries and overlaid real-time guidance messages such as "Stay on the Lane." In well-lit environments, the detection accuracy remained consistent, ensuring reliable lane tracking.

However, certain limitations were observed in non-ideal conditions:

- **Adverse Weather Conditions:** In heavy rain and fog, visibility of lane markings was reduced, leading to occasional misdetections. Faded or partially obscured lanes were sometimes misclassified as road background.
- **Lane Occlusion:** When lane markings were obstructed by vehicles, detection accuracy decreased. The system occasionally failed to maintain continuity in partially visible lanes.
- **Curved and Dashed Markings:** While the system performed well on straight roads, minor inaccuracies were noticed on sharp curves and broken lane markings.

Despite these challenges, the lane detection module remained effective in standard driving environments, ensuring robust tracking even in moderate traffic conditions.



Fig. 4. Lane and object detection results: The system successfully identifies lane boundaries and vehicles, providing real-time alerts.

B. Object Detection Performance

The object detection module, implemented using YOLOv3, achieved an average precision of 85% for vehicle detection. The model successfully identified multiple vehicles and provided dynamic warnings based on surrounding traffic conditions. It performed well in both normal and congested traffic scenarios, ensuring real-time responsiveness.

However, certain limitations were noted:

- **Pedestrian Detection:** The system detected pedestrians in unobstructed conditions but exhibited lower detection rates in complex environments. This was likely due to dataset bias, as the training data primarily contained vehicle images.
- **Static Object Misclassification:** Some road signs and stationary objects were occasionally misclassified as vehicles, particularly in cluttered urban settings.
- **High-Speed Motion Blur:** At higher vehicle speeds, detection accuracy decreased due to motion blur and reduced detection windows.

A key advantage of the proposed system is its real-time performance. It operated at **95–100 FPS**, ensuring smooth execution across various driving speeds while maintaining a balance between accuracy and efficiency.

C. Comparative Performance Analysis

To evaluate the system's effectiveness, we compared its performance with conventional lane detection methods. Table III presents a detailed comparison in terms of accuracy and processing speed (FPS). The evaluation revealed that Canny Edge Detection achieved the highest FPS (200) but lacked robustness in complex environments, while the Hough Transform, though moderately accurate (80.5%), was computationally expensive. LaneNet and SCNN generated the most precise lane detection but suffered greatly diminished FPS, limiting their real-time applicability. YOLOv3 performed best in object detection at 100 FPS but needed lots of computational power. The proposed system adequately balanced accuracy and real-time performance with competitive lane detection accuracy (90.0%) while running at 95-100 FPS, and thus well-suited for autonomous driving use.

The outcome means that the given system successfully combines lane and object detection, with uniform performance under standard driving conditions. Even though the system had difficulty in adverse weather and high-speed scenarios, it displayed substantial superiority to the conventional edge-based methods. The integration of YOLOv3 with lane detec-

TABLE III
PERFORMANCE COMPARISON OF LANE AND OBJECT DETECTION METHODS

Method	Accuracy (%)	FPS
LaneNet	96.38	50
SCNN	96.53	7.5
Canny Edge Detection	85.0	200
Hough Transform	80.5	100
YOLOv3 (Object Detection)	85.0	100
CNN (Generic)	92.8–98.96	Varies
Integrated (YOLOv3 + Lane Detection)	90.0 (lane) / 85.0 (objects)	95–100

tion gets the optimum trade-off between accuracy and real-time processing, making it appropriate for autonomous driving applications.

V. CONCLUSION

This project presents an end-to-end system that integrates lane detection and object detection to enhance the performance, safety, and situational awareness of autonomous vehicles. The methodology suggested is a mixture of conventional computer vision techniques, such as edge detection and the Hough Line Transform, and deep learning-based object detection using YOLOv3. This offers a robust balance between accuracy and real-time processing and is thus perfectly suited for real-time autonomous driving systems. The metrics of evaluation demonstrate the robustness of the system with high accuracy in lane and object detection and a stable frame rate to ensure real-time responsiveness.

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